

YULIA VOLKOVA

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I am a MATS 9.1 scholar working on AI control evaluations with Rhys Ward and Redwood Research. I have 7+ years of experience building insider threat detection and communication surveillance systems at Behavox for Tier-1 financial institutions. I first hand witnessed AI integration into the regulation sector, which inspired me to move and work on AI safety. Further details can be found on my [website](#).

EDUCATION

London School of Economics

MSc Economics

- Distinction ([dissertation in behavioural economics](#))
- Courses included Advanced Macroeconomics, Econometrics, Quantitative Methods

London School of Economics

BSc Philosophy & Economics

- Distinction
- Courses included Mathematical Methods, Statistics, Econometrics, Philosophy of Science

SELECTED PUBLICATIONS

2026 Control Evaluations for Covert Malicious Fine-Tuning. *Submitted to NeurIPS 2026*
Buckworth, T., Moreno, R., Panda, A., **Volkova, Y.** Supervised by Dr Francis Rhys Ward.

[Residual Stream Decoders for Probing Model Activations.](#) *Submitted to ICML*
Algoverse AI Safety Research Fellowship. PI: Callum McDougall, mentor: Nicky P.

POSITIONS AND PROGRAMS

[MATS 9.1 Extension](#) *Apr 2026 – Present*

- Continuing control evaluations research under Francis Rhys Ward

[MATS 9.0](#) *Jan – Apr 2026*

- Control evaluations for covert malicious fine-tuning under Francis Rhys Ward, advised by Tyler Tracy and James Lucassen (Redwood Research)
- Selected for MATS 9.0 Spotlight Talk — opened the London event

[Athena Women in AI Safety Research Fellowship](#) *Nov 2025 – Jan 2026*

- [Mechanistic interpretability approach to monitoring chain-of-thought faithfulness](#) under Arun Jose

- Gathered feedback from Adam Karvonen, Goodfire team, and Uzay Macar; integrating in MATS 9.1 extension phase

SPAR Fellowship

Oct – Dec 2025

- CoT monitorability project under Qiyao Wei (MATS 8.0, Cambridge University)

Visiting Member, London Initiative for Safe AI (LISA)

Sept 2025 – Present

Algoverse AI Safety Research Fellowship

Jul – Aug 2025

- Mechanistic interpretability under Nicky P
- Cooperative AI under Max Kleiman-Weiner

ARBOx2 Oxford AI Safety Bootcamp

Jul 2025

- Completed the ARENA curriculum

MEDIA & TALKS

- **MATS 9.0 Spotlight Talk** — opened the London event ([YouTube](#))
- **FellowCast Podcast at Constellation** — "Covert Malicious Fine-Tuning" ([Spotify](#))

WORK EXPERIENCE

Behavox

2017 – 2025

Insider threat detection, communication surveillance, and compliance

Data Science Production Team Lead

2024 – 2025

- Led development and deployment of LLM-driven Java applications across 3 products and 14 languages, using a pipeline of fine-tuned models (SBERT → DeBERTa → Qwen-14B)
- Managed execution of Java applications on the Apache Beam Dataflow runner; monitored GPU utilization and used BigQuery datasets to debug pipeline behavior and resolve incidents
- Led cross-team roadmapping for ML features under production constraints
- Built reusable tooling—CI/CD templates and MLflow tracking—enabling rapid MVPs for client deliveries

Insider Threat Product Lead

2020 – 2024

- Designed risk taxonomy and classifiers for insider threats—data exfiltration, disgruntlement, flight risk
- Worked directly with heads of insider risk at Citadel, SoftBank, and other Tier-1 financial institutions to validate detection models against real threat landscapes
- Led R&D and production of high-recall (81% at 0.5% FPR) misconduct detection systems using RoBERTa + SBERT pipeline
- Validated that shorter text sequences in RoBERTa training maintain performance, doubling training speed and halving costs
- Built internal tooling (chatbot, recommendation system) serving 200+ daily users

Data Scientist

2017 – 2020

- Landed proof of concepts that resulted in 3+ years client contracts
- Improved compliance model recalls from 70% to 90%
- Reduced client bug requests from 1/day to 1/month by setting up an efficient QA process
- One of the first Data Science hires; created foundational documentation during company scaling

ADDITIONAL SKILLS AND INTERESTS

Programming Languages: Python, PyTorch, HuggingFace, Pandas, NumPy, Scikit-learn, FastAPI, spaCy, SQL, Java, Kotlin

Machine Learning & NLP: LLM fine-tuning (LoRA/qLoRA), prompt engineering, transformers, SBERT, PCA, t-SNE, Weights & Biases

Infrastructure: AWS S3, GCP, Docker, Nomad, Kubernetes, TeamCity, Gitlab CI/CD pipelines

Statistics & Econometrics: Hypothesis testing, time-series analysis, A/B testing, IV estimation, panel regression

Leadership: Mentorship, technical interviewing, product roadmapping, cross-team communication

Other Interests: Oil painting, skiing, hiking